

Manual

HYDRA Interfacing Module to SAP PM (CC3) SAP-PMCC 8.2

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1 HYDRA Interfacing Module to SAP R/3 PM (CC3)

Overview

Purpose

You can use the interfacing module SAP PM (CC3) to download maintenance and service orders for an operation from SAP PM. In HYDRA, you can record postings for these operations and upload the resulting times to SAP PM. You use the CA-PDC / CC3 interface to connect to SAP PM.

Implementation notes

You use the function package for the following purposes:

- You use SAP PM to plan maintenances and you want to use the MES to record the times for these orders.
- You use SAP CS (Customer Service) to plan and run services and you want to record times for the service orders via MES.

Integration

If you use the component SAP-PMCC3, the orders/operations transferred with this component are used for a great number of further postings in HYDRA.

Features

- Transfer of order data
 - Download of the released operations from PM and integration as HYDRA maintenance orders/operations (initial and delta download and delete function)
- Recording of times
 - Recording of start/interruption/end of processing and posting of the times according to the HYDRA configuration and validation checks
- Upload of operation postings
 - Upload to SAP PM of the actual times (performances) recorded for a specific SAP time ticket
- Interface configuration
 - Configurable assignment of the times recorded in HYDRA resource performance accounts to the SAP activity types

2 Mapping of the SAP-PMCC3 in HYDRA

Summary

In the course of a connection of HYDRA to SAP PM, HYDRA must collect PM-relevant data and transfer them to SAP. The base data here are PM-maintenance and service orders transferred from SAP to HYDRA.

The download trigger for PM-maintenance orders (PP-PDC / CC3) comes from SAP. The data are transferred as IDoc (intermediate document) and maintained in HYDRA.

The upload of the confirmations for PM maintenance orders is controlled via SAP in accordance with the requirements specified by the user.

To realize the communication with the BDE subsystems, SAP provides several IDocs via the PP-PDC interface. The following IDocs are used:

Download of PM maintenance orders (PP-PDC/ CC3):

IDoc type:	OPERA3
Message type:	OPERA3
Message function:	APP/ DEL/ UPD
Segment type:	OPERA3

Download of PM upload requests (PP-PDC/ CC3):

IDoc type:	REQUI3
Message type:	REQUI3
Message function:	REQUI3
Segment type:	REQUI3

Upload of confirmations of maintenance orders (PP-PDC/ CC3):

IDoc type:	CONF32
Message type:	CONF32
Segment type:	CONF5

Download of operation data SAP → HYDRA

The operations are transferred in an IDoc of the *OPERA3* type. This may be an initial, delta or deletion download.

The upload request is transferred in an IDoc of the *REQUI3* type. When this is received in HYDRA, confirmations that exist already in HYDRA interface tables will be transferred to SAP.

Upload of confirmations HYDRA → SAP

Confirmations are uploaded either cyclically from HYDRA or from SAP R/3. The interface offers numerous options to this end so that the specific requirements can be mapped.

The transfer of confirmations/uploads to SAP R/3 is either controlled by HYDRA or by SAP R/3 and is made in an IDoc of the *CONF32* type.

3 Behavior Depending on the Transfer Types

Initial download behavior

An initial download of the order and operation data will only take place when the complete system is commissioned and a first database is to be created in HYDRA. In addition, also other scenarios are possible by which the entire operations base will be replaced.

During an initial download all operations from SAP will be transferred that have at least the status "released" or that are not technically completed yet. This means that also (end) confirmed operations will be transferred but not imported in HYDRA.

When the initial download is received, the current operations data base in HYDRA will be deleted and be replaced by the new operations data. The current data will immediately be available.

An initial download will not only delete the order base existing in HYDRA but also all current times and quantities entered for orders and/or transactions.



In addition, all operations that are being executed will not be deleted. This means that they must be interrupted and/or terminated manually in HYDRA.

Conclusion: Any initial download during the operation of HYDRA must be used with greatest care and it would be useful to contact the MPDV Support beforehand.

For security reasons, the initial download function needs to be enabled explicitly as of the below-mentioned program version. Activation is performed by an INI configuration.

.lib\b_anr.dll

V8.1.1.326

Delta download behavior

A delta download creates new operations in HYDRA that will then be added to the database. Another function of the delta download is the modification of already transferred operations. Modifications of operations with the status "Running", "Finished" or "Deleted" are not allowed by the HYDRA default settings.



CUSTOMIZING information: Modifications of the operations depend on whether the flag "alterable order data" is set for the corresponding status in the status configuration.

If an operation cannot be modified to ensure the consistency in HYDRA (in general for running operations), this will be logged in HYDRA and be saved for error tracking purposes.

The basis for the delta download in SAP is the database table ORDCOM.

Deletion download behavior

A deletion download deletes those operations that are no longer necessary in the production process from the HYDRA database.

As with the modifications, the deletion result depends on the operation's status. Operations identified as "Running", "Interrupted", "Finished" or "Deleted", will not be deleted.

Moreover, the confirmation number (CONF_NO) will be checked to identify the orders via order, sequence and transaction. This prevents any accidental deletion of orders that are seemingly the same but which have different confirmation numbers.

Behavior when re-importing master data in SAP

When master data for an existing and released order that has already been transferred to HYDRA are re-imported in SAP this impacts also the interface to HYDRA.

When master data are re-imported, SAP will assign new confirmation numbers for the individual operations even though the order and operation numbers won't change. These are then transferred together with the next delta download to HYDRA. The confirmation/upload number will then be updated in HYDRA.



The update will also be made when the current status is "Running". In this case, however, ONLY the confirmation/upload no. will be changed and all other data (order quantity, scheduling, etc.) will not be updated.

Behavior during the technical completion in SAP

If an order is technically completed in SAP this will lead to a deletion download at the interface. This means that the order data in HYDRA will be deleted.

4 Download of Operation Data SAP --> HYDRA

Summary

Transaction data

The following structure is part of the PP-PDC/ CC3 interface and is used to transfer maintenance orders to sub-systems.

Message type: OPERA3

IDoc type: OPERA3

Segments: E2OPERA1

Message function:

APP	→	Delta download
DEL	→	Deletion download
UPD	→	Initial download

Field name	T	L	Description	Rel. KK3	Usage in HYDRA
RUECK*	NUMC	10	Confirmation number of the operation	x	Saved for the confirmation, besides no other usage
AUFNR	CHAR	12	Order number	x	Order number
APLFL	CHAR	6	Operation sequence	x	Not used
VORNr	CHAR	4	Operation number	x	Operation number
UVORN	CHAR	4	Sub-operation	x	Sub-operation no.
SPLIT*	NUMC	3	Split number	x	Not used
KAPAR*	CHAR	3	Capacity type	x	Not used
BDEGR*	CHAR	3	Grouping subsystem connection	x	Not used
MGVRG	DEC	10.3	Default quantity	-	
ASVRG	DEC	10.3	Scrap quantities	-	
MEINH	CHAR	3	Operation quantity unit	-	
UMREN	DEC	5.0	Denominator for the conversion	-	
UMREZ	DEC	5.0	Numerator for the conversion	-	
KMEIN	CHAR	3	Header quantity unit	-	
UNTMG	DEC	10.3	Underdelivery quantity	-	
UEBMG	DEC	10.3	Overdelivery quantity	-	
ACTI1	DEC	10.3	Planned activity 1	-	
UNIT1	CHAR	3	Unit of the planned activity 1	-	

Field name	T	L	Description	Rel. KK3	Usage in HYDRA
ACTI2	DEC	10.3	Planned activity 2	-	
UNIT2	CHAR	3	Unit of the planned activity 2	-	
ACTI3	DEC	10.3	Planned activity 3	-	
UNIT3	CHAR	3	Unit of the planned activity 3	-	
ACTI4	DEC	10.3	Planned activity 4	-	
UNIT4	CHAR	3	Unit of the planned activity 4	-	
ACTI5	DEC	10.3	Planned activity 5	-	
UNIT5	CHAR	3	Unit of the planned activity 5	-	
ACTI6	DEC	10.3	Planned activity 6	-	
UNIT6	CHAR	3	Unit of the planned activity 6	-	
LMNGA	DEC	10.3	Planned yield	-	
XMNGA	DEC	10.3	Planned scrap quantity	-	
ISTAT	CHAR	5	Status of the operation	x	
ISM01	DEC	10.3	Actual activity 1	-	
ISM02	DEC	10.3	Actual activity 2	-	
ISM03	DEC	10.3	Actual activity 3	-	
ISM04	DEC	10.3	Actual activity 4	-	
ISM05	DEC	10.3	Actual activity 5	-	
ISM06	DEC	10.3	Actual activity 6	-	
LEK01	CHAR	1	End indicator for activity 1	-	
LEK02	CHAR	1	End indicator for activity 2	-	
LEK03	CHAR	1	End indicator for activity 3	-	
LEK04	CHAR	1	End indicator for activity 4.	-	
LEK05	CHAR	1	End indicator for activity 5	-	
LEK06	CHAR	1	End indicator for activity 6	-	
ARBPL	CHAR	8	Workplace	x	HYDRA workplace
WERKS	CHAR	4	Plant for the workplace	x	Plant
ARBPI	CHAR	8	Actual workplace	x	Not used
WERKI	CHAR	4	Plant for the actual workplace	x	Not used
ISMNW	DEC	10.3	Actual working time (must not be neg.)	x	Not used
ISMNE	CHAR	3	Unit of the actual work	x	Not used
ARBEI	DEC	6.1	Planned work	x	Target BMK11 *)
ARBEH	CHAR	3	Unit of the planned work	x	S/SEC/MIN/STD/H/HUR/HR
OFMNW	DEC	6.1	Remaining work (must not be neg.)	x	Not used
OFMNE	CHAR	3	Unit of the remaining work	x	Not used
LEKNW	CHAR	1	Indicator: no remaining work	x	Not used
FSAVD	DATS	8	Earliest start date	x	PPS start date
FSAVZ	TIMS	6	Earliest start time	x	PPS start time
SSEDD	DATS	8	Latest end date	x	PPS end date

Field name	T	L	Description	Rel. KK3	Usage in HYDRA
SSEDZ	TIMS	6	Latest end time	x	PPS end time

*) The actual working time (field ISMNW) must not exceed a duration of 590 hours.

Upload request

The upload request can be used to control the upload of confirmations to SAP. If such an upload request is received from SAP in HYDRA, all available confirmations will - in turn - be transferred to SAP.

Message type: REQUI3
IDoc type: REQUI3
Segments: REQUI3
Message function: REQ → Upload request

5 Upload of Confirmations HYDRA --> SAP

Summary

Confirmation PP-PDC/ CC3

Record type	SAP meaning	HYDRA action to be triggered
I20	Time ticket - partial end	Automatic or manual order interruption at the BDE terminal or at the BDE console
I40	Time ticket end	Notification of the end of an order at the BDE terminal or at the BDE console

HYDRA BDE confirms to SAP on the basis of time tickets. Transferred are the staff durations from the B-records (personnel postings). They are transferred to SAP together with the personnel number.

A transfer of correction notifications is **not** included in the SAP default interface PP-PDC/ CC3.

Confirmation structure "E2CONF32"

Message type: CONF32
IDoc type: CONF32
Segments: E2CONF5

Field name	Type	Length	Text	Usage in HYDRA		
Record type	CHAR	3	Record type <i>of the confirmation</i>	I20 or I40		
TERID	CHAR	4	Terminal ID	Not used		
LDATE	DATS	8	Logical date/ <i>actual date of the confirmation</i>	Logoff date		
LTIME	TIMS	6	Logical time/ <i>actual time of the confirmation</i>	Logoff time		

Field name	Type	Length	Text	Usage in HYDRA		
ERDAT	DATS	8	Entry date <i>of the confirmation</i>	Date of the confirmation		
ERTIM	TIMS	6	Entry time <i>of the confirmation</i>	Time of the confirmation		
BUDAT	DATS	8	Posting date <i>of the confirmation</i>	set according to shift date of the posting record		
ARBPL	CHAR	8	Workplace	HYDRA workplace		
WERKS	CHAR	4	Plant	Plant		
ZAUSW	NUMC	8	<i>Badge number</i>	<i>Time ID number</i>		
AUFNR	CHAR	12	Order number	according to specifications		
VORNr	CHAR	4	Operation number	according to specifications		
UVORN	CHAR	4	Sub-operation <i>number</i>	according to specifications		
SPLIT	NUMC	3	<i>Split number</i>	Not used		
KAPAR	CHAR	3	Capacity type	Not used		
PEDD	DATS	8	Predicted end date	Not used		
PEDZ	TIMS	6	Predicted duration end time	Not used		
LEKNW	CHAR	1	Indicator: no remaining work	Not used		
LTXA1	CHAR	40	Upload text	Not used		
ISMNW *	DEC	6.1	Actual work	P_DAUER		
ISMNE	CHAR	3	Unit of the actual work	Hrs/ min/ sec*		
LEARR	CHAR	6	Activity type	Activity type of the actual workplace		
IDAUR *	DEC	4.1	Actual duration	RPA 11		
IDAUE	CHAR	3	Unit of the actual duration	Hrs/ min/ sec*		
ODAUr *	DEC	4.1	Remaining duration	Not used		
ODAUE	CHAR	3	Unit of the remaining duration	Not used		
OFMNW *	DEC	6.1	Remaining work	Not used		
OFMNE	CHAR	3	Unit of the remaining work	Not used		
ISDD	DATS	8	Start date for execution	Login date		
ISDZ	TIMS	6	Start time for execution	Login time		
IEDD	DATS	8	End time for execution	Logoff date		
IEDZ	TIMS	6	End time for execution	Logoff time		

6 Activity Types SAP

Overview

HYDRA menu	System administration → MES Link Enabling (MLE) → Activity types SAP
FEDRA menu	System administration → MES Link Enabling (MLE) → Activity types SAP
Transaktionscode	mlecas
Function authorization	mlecas.*

Purpose

You use this setting if you want to link the activity types required for uploads to SAP PM with the workplaces created in the system.

You edit the assignment in relation to a workplace known in the system and the target module in SAP.

Integration

The setting is used for uploads to SAP PM.

Requirements

You have created workplaces and groups in the system.

Field descriptions

Activity type

Activity type to stipulate how to upload to SAP.

Cost center

Not relevant

SAP_Modul

Enter "PM" permanently for uploads to SAP PM.

Name

Free text

Year

No functional use: Date of validity

Plant

No functional use: Plant where the workplace is located

Workplace

Workplace to which the setting applies

From PPS

Don't set the identification.

7 MYERPRCK - Program Parameters

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Purpose

Use the upload program myerprck.exe/out to create confirmations/uploads to higher-level systems. In addition to the settings you make directly in the applications, you can also use program parameters to control confirmations/uploads.

Integration

The confirmation/upload is integrated with numerous components, for example:

- Shop floor data collection
- Tracking and tracing as well as material and production logistics
- Detailed scheduling

Available program parameters:

Parameters	Meaning/use	Relevant interfaces	Productive release
Program parameters to control processing:			
/MESTYP=XXXX	The parameter MESTYP defines the structure to be generated.	All	Yes
/GRP=XXXX	The grouping type specifies the criterion by which uploads should be grouped. Possible values: PLANT --> Groups by plant	Requires customizations	Requires customizations
/V=sssss	Since SAP R/3 PP does not support correction postings, HYDRA allows to retain confirmations/uploads for correction purposes in HYDRA for a specific period of time. Use the parameter /V=sssss (sssss =	EIS-ERP EIS-XPPS SAP-PPDC SAP-PPREM	Yes

Parameters	Meaning/use	Relevant interfaces	Productive release
	delay time in seconds) to activate the above described delay when the upload program is called. Examples: <pre>myerprck.exe/out /V=3600</pre> The system only uploads postings that are older than one hour.	SAP-PPPI SAP-PMCC3 SAP-PSCC4 SAP-COILV	
/BIS=DDMMYYHHMM /BIS=HHMM /TILLDATE=MM/DD/YYYY /TILLTIME=sec after midnight	Use the parameter /BIS=DDMMYYHHMM (date + time) when calling the upload program to enter the delay as a point in time. You can enter this point in time with date and time or you can just enter the time in the format "HHMM". In the latter case, the time refers to the current day. <pre>Myerprck.exe /BIS=2505110600</pre> This parameter uploads postings that were recorded until 06:00 a.m. on 25 May 2011. <pre>Myerprck.exe /BIS=0600</pre> This parameter uploads postings that were recorded until 06:00 a.m. of the current day.	EIS-ERP EIS-XPPS SAP-PPDC SAP-PPREM SAP-PPPI SAP-PMCC3 SAP-PSCC4 SAP-COILV	Yes
/TZ=+/-sssss	Use the parameter /TZ=+/-sssss to adapt uploads to different time zones. The parameter adjusts the time specifications entered in the fields EXEC__START_TIME, EXEC_FIN_TIME and LOGTIME of the upload structure of the SAP-PPDC interface according to its specifications.	SAP-PPDC	Yes

Parameters	Meaning/use	Relevant interfaces	Productive release
/KST=XXX	Use this parameter to restrict the data to be uploaded. In this case, the system only uploads data of a specified cost center. Use the parameter /KST=XXX (XXX = cost center, a max. of 8 characters) when calling the upload program myerprck.exe/out to enable the above-described restriction. Then the system only uploads data records that were posted to machines of the specified cost center. The system checks the cost center of the machine/workplace that is entered as the posting workplace/machine in the posting record. The system only checks the cost center of the workplace/machine. You can specify the parameter several times per call. <u>Example:</u> <pre>Myerprck.exe /KST=BDE100 /KST=BDE200</pre> The system only uploads records that were posted onto machines of the cost center BDE100/BDE200.	EIS-ERP ESI-XPPS SAP-PPDC SAP-PPREM SAP-PPPI SAP-PMCC3 SAP-PSCC4 SAP-COILV	Yes
/CLEAR_RES	Use the parameter "/CLEAR_RES" to assign an "X" to the field CLEAR_RES of the upload structure when it comes to a final confirmation/upload (record type L40). Consequently, SAP will clear open reservations for the respective order.	SAP-PPDC	Yes
/NEG_MENGE	By default, quantities (L20/L40) cannot	SAP-PPDC	Yes

Parameters	Meaning/use	Relevant interfaces	Productive release
	be uploaded to SAP PP using partial confirmations/uploads via the SAP-PPDC interface if data is collected at the same time via the total quantity counter of MDE machines, since SAP is not able to process negative quantities. This type of collection can result in negative quantity postings for yield when OPs are finished. This restriction does no longer apply, if it is possible to process such negative postings (e.g. by using the SAP standard BAPI or customizations). In this case, you can use the program parameter /NEG_MENGE to enable the upload of these quantities.		
/LA_MNR	The SAP_PMCC3 interface requires the activity type to be uploaded to SAP PM. The activity type can be identified via the machine/workplace where the posting was performed. Use this program parameter to enable identification of the activity type. Then the system uses the machine to identify the activity type from the activity types kept in HYDRA.	SAP-PMCC3	Yes
/IDENT_PRAEFIX=	In the upload structure of the SAP-PPDC interface, the field EX_IDENT uniquely identifies uploads from subsystems. HYDRA populates the field. You can add a prefix to the EX_IDENT field to differentiate between uploads	SAP-PPDC SAP-PPDCC	Yes

Parameters	Meaning/use	Relevant interfaces	Productive release
	from various HYDRA subsystems connected to one SAP instance. <u>Example:</u> <pre>Myerprck.exe /IDENT_PRAEFIX=ABC</pre> The prefix may only include hexadecimal characters: A –H und 0 – 9.		
/ABZEICH=XX	While customizing the order type, you can specify that only signed data records are uploaded. Use the parameter /ABZEICH=XX to specify a period of time in days after that you can upload even unsigned data records.	EIS-ERP EIS-XPPS SAP-PPDC SAP-PPREM SAP-PPPI SAP-PMCC3 SAP-PSCC4 SAP-COILV	Yes
/TRANSFER=	Use the parameter "/TRANSFER=" to only upload records whose specifications were transferred from a specific system. The transfer indicator is set during HYDRA inbound processing and may vary from interface to interface.	EIS-ERP ESI-XPPS SAP-PPDC SAP-PPREM SAP-PPPI SAP-PMCC3 SAP-PSCC4 SAP-COILV	Yes

Parameters	Meaning/use	Relevant interfaces	Productive release
/NOTTRANSFER=XXX	Use the parameter "/NOTTRANSFER=" to only upload records whose specifications were NOT transferred from a specific system. The transfer indicator is set during HYDRA inbound processing and may vary from interface to interface.	EIS-ERP ESI-XPPS SAP-PPDC SAP-PPREM SAP-PPPI SAP-PMCC3 SAP-PSCC4 SAP-COILV	Yes
/SEK	The EIS-ERP interface uploads the times of resource performance accounts in hours. In particular with very short lead times this may effect that logon times are cut off by a conversion into hours. Use this program parameter to upload times in seconds.	EIS-ERP ESI-XPPS	Yes
/RMTYP=	When customizing the order type, you can assign an upload type to the order type. Use this program parameter to only upload data records of this upload type. You can specify the parameter several times per call.	EIS-ERP ESI-XPPS SAP-PPDC SAP-PPREM SAP-PPPI SAP-PMCC3 SAP-PSCC4 SAP-COILV	Yes

Parameters	Meaning/use	Relevant interfaces	Productive release
/KAT=	When customizing the order type, you can connect the order type with a category. Use the program parameter /KAT= to only upload data records of this category. You can specify the parameter several times per call.	EIS-ERP ESI-XPPS SAP-PPDC SAP-PPREM SAP-PPPI SAP-PMCC3 SAP-PSCC4 SAP-COILV	Yes
/SART=	The system only uploads ADE log postings of the specified record type. Therefore, you can use different program parameters per call and record type for uploading. Requirement: You have to activate the corresponding uploads when customizing the order type. You can specify the parameter several times per call. <u>Example:</u> <pre>Myerprck.exe /SART=A /SART=E</pre> ⇒ The system only uploads A and E records.	EIS-ERP ESI-XPPS SAP-PPDC SAP-PPREM SAP-PPPI SAP-PMCC3 SAP-PSCC4 SAP-COILV	Yes
/NOLOCK	When starting the upload program, the system checks if there are any lock entries for the database table ADE_PROTOKOLL. If this is the case,	All	Requires customizations

Parameters	Meaning/use	Relevant interfaces	Productive release
	the upload is not carried out. You can use this program parameter to prevent this check. Set this parameter, in particular, if the upload is not based on the table ade_protokoll.		
/EINH_CC34	The interfaces SAP-PMCC3 and SAP-PSCC4 transfer the uploaded activity quantity in seconds (SEC) to SAP. Use the parameter "/EINH_CC34" to upload the data in other units. The following units are supported: Hours: H, HUR, STD Minutes: MIN Seconds: SEC <u>Example:</u> Myerprck.exe /EINH_CC34=HUR The system uploads the recorded times in the unit "HUR" (hours).	SAP-PMCC3 SAP-PSCC4	Yes
/SDAT_STORNO	The SAP-PPPDCC interface transfers the change date along with the correction records. Use this program parameter to upload the initially collected shift date instead.	SAP-PPPDCC	Yes
/NORFC_STORNO	The SAP-PPPDCC interface transfers the cancellation records via sRFC. Use the program parameter to transfer the data in the IDoc format to SAP. To do	SAP-PPPDCC	Yes

Parameters	Meaning/use	Relevant interfaces	Productive release
	so, inbound processing must be implemented in SAP. The system uploads the cancellation records via the standard PP-PDC segment (with record type K20/K40) as if the PP-PDCC license was not available.		
/PI	If you use the SAP Process Integration (previously: Exchange Infrastructure) to communicate with SAP, the version of the transferred segment is checked more strictly. Use the program parameter to transfer segment names with the version number (i.e. the trailing zeros of the segment name).	SAP-PPDC SAP-PMCC3 SAP-PSCC4	Yes
/INDEX_TMP_TABLE	Use this parameter to accelerate uploads if ORACLE is used as database system and large amounts of data are affected. To do so, use an index for a temporary table where all data to be uploaded is transferred in a first step.	All	Requires customizations
/UE_PARAMS=	Program parameter for the stand-alone user exit processing (DD format).	Various	Yes
/NOSTORNO	Use this program parameter to prevent cancellation records from being uploaded. Therefore, you can use different program parameters per call and record type for uploading. Requirement: You have to activate the	All	Yes

Parameters	Meaning/use	Relevant interfaces	Productive release
	corresponding uploads when customizing the order type.		
/RECALC_NEG_YIELD	Use this parameter to offset negative yield with already posted positive uploads.	SAP-PPPDCC	Requires customizations
Program parameters to use the SIGUSR communication:			
/LOGGING	Use this program parameter to activate communication from the database table HYD_LOGGING. To do so, a customization might be required.	INDIVIDUAL CASE	Yes
/WAIT_SIGUSR1=XX	The program parameter specifies the time in seconds that has to pass before the upload is performed via the SIGUSR communication even without trigger.	INDIVIDUAL CASE	Yes
/PEEK_SIGUSR1=XX	Use this parameter to delay execution of an action triggered by the SIGUSR communication. The delay time is entered in seconds for this parameter. The program interprets this time as follows: If within the next second after the initial trigger there is another trigger, then wait for not more than <specified value> seconds. If in a specific case, triggers would indeed arrive every second then the	INDIVIDUAL CASE	Yes

Parameters	Meaning/use	Relevant interfaces	Productive release
	WAIT_SIGUSR time (e.g. 120 seconds) would apply; i.e. the system would in fact perform the upload after 2 minutes.		
/SEND_SIGUSR1=	This program parameter defines which other process/ program must be triggered after processing by the SIGUSR communication. Specify the process/program WITHOUT file extension.	INDIVIDUAL CASE	Yes
/COUNT_SIGUSR1=XX	Uploading in signal mode can hardly be subjected to tracing. This is due to the fact that the program in those cases is started once via the scheduler but won't shut off. Any redirection of the program call with -d to a log file will then necessarily lead to very large log files, which will negatively affect the performance. Use the new program parameter /COUNT_SIGUSR1=XX to specify after how many calls the program will automatically shut down. A call in these instances is both, a call via SIGUSR communication and the cyclical program execution which is controlled via the parameter /WAIT_SIGUSR1. Then the scheduler restarts the program. But this will lead to a time period "t" during which SIGUSR calls will not be processed. It is, however, assumed that this will not lead to data losses since the data to be uploaded are already saved to	INDIVIDUAL CASE	Yes

Parameters	Meaning/use	Relevant interfaces	Productive release
	the DB. Benefits: If the program is started via a script (*.scr) from the scheduler, you can store there the routine to generate a date/ time stamp file name for the log file to be created. This allows to restrict the log file size.		
Program parameters for debugging/ tracing/ testing/ logging purposes:			
/ONLYERR	This program parameter specifies that system log entries are only created if an error occurred during uploading. This reduces the entries in the system log.	All	Yes
/SIM	The system does not upload/confirm data during simulations (the uploaded/confirmed indicator is set to "True").	All	No
/SIMULATION	The system does not upload/confirm data to SAP during simulation (confirmed/uploaded indicator will not be changed).	All	No

8 Application-Relevant Settings in HYDRA

Maintenance of the HYDRA distribution model - inbound processing

Use the [HYDRA distribution model](#) to maintain entries for HYDRA inbound processing:

Name of the parameter	Value
To process PM/ CS production orders	
Message type	OPERA3
Priority	None
Command	mle72imp.scr
Command parameter	/VARIANT= <MLE variant to be used>
Description	PM-CC3 – download PM/ CS orders
Log. target system	Created logical system
Storage duration	10
To process the upload request	
Message type	REQUI3
Priority	High
Command	hysapupl.scr
Command parameter	/UPLSEGNAM=E2CONF5
Description	PP-CC3 – Upload request
Log. target system	Created logical system
Storage duration	10

Maintenance of the HYDRA distribution model - outbound processing

Use the [HYDRA distribution model](#) to maintain entries for HYDRA outbound processing:

Name of the parameter	Value
To upload time tickets	
Message type	CONF32
Description	PP-CC3 – Upload time tickets
IDoc-Typ	CONF32
Storage duration	10
Log. target system	Created logical system
Segment name 1	E2CONF5

Activation of initial download

As of program version

.lib\b_anr.dll V8.1.1.326

the initial download has to be enabled explicitly for security reasons.

Create the following entry in [HYDRA INI configuration](#) if you would like to activate the initial download for the system:

Parameter name	Value
INI name	SAP
Section	INITIAL_DOWNLOAD_ACTIVATION
Key	ACTIV_TILL
Value	<date value in the format MM/DD/YYYY>
Active	Yes
Comment	Activation of initial download

9 Application-Relevant Settings in SAP

Customizing the order type

In SAP the CA-PDC interface (KK3) will only take those maintenance and service orders into account for which the order type has been marked as “BDE-active”. This is marked in Customizing (OIOE).

For each relevant combination of plant and order type the indicator “BDE-active” must be set.

Maintenance at the workplace

Once an order type is identified as “BDE-active” the CA-PDC interface (KK3) will only take those transactions into account for which at least one subsystem grouping is saved to the workplace.

The subsystem grouping at the workplace is maintained using the workplace maintenance (CR02/ IR02) → basic data → subsystems. There, the relevant subsystem can be selected from several saved subsystems.

Definition of new subsystem groupings

To the extent that the subsystem groupings included in the SAP delivery do not suffice, it is possible to define new ones using SAP Customizing - *SPRO* → *Personnel time management* → *Business data collection* → *General settings* → *Define grouping for subsystem connection*.

Setting of the posting times

Depending on the settings in SAP, the CA-PDC interface (KK3) supports two confirmation scenarios:

- Immediate posting

If the “Immediate posting” indicator is active in Customizing (CI31), HYDRA will immediately post time ticket confirmations transferred to SAP. If this posting cannot be made - if for example a maintenance or service order is being blocked - the confirmations will stay prebooked and will be posted during the next posting run.

- Posting using Job

If the “Immediate posting” indicator is not set in Customizing (CI31), the confirmations will be prebooked. They will then be posted later depending on the job, using Job CIP3.

Planning of relevant jobs

The following programs/ reports must be planned as job to ensure that the PP-PDC interface will operate automatically:

Program/ Report	Meaning	Please note:
CIBDOPDE	Download maintenance/ service orders	Planning with one variant
SAPCDUP3	Download of the upload request	Planning with one variant

Maintenance of the SAP partner agreement/profile – outbound processing

Name of the parameter	Value
To download PM/ CS orders	
Partner number	Created logical system
Partner type	LS
Message type	OPERA3
Message function	APP/ UPD/ DEL
Receiver port	Created port
Package size	1
Output mode	Transmit IDoc immediately
Basis type	OPERA3
To download the upload request	
Partner number	Created logical system
Partner type	LS
Message type	REQUI3

Name of the parameter	Value
Message function	REQ
Receiver port	Created port
Package size	1
Output mode	Transmit IDoc immediately
Basis type	REQUI3

Maintenance of the SAP partner agreement/profile – inbound processing

Maintain the following settings for inbound processing in the partner agreement/profile in SAP (WE20)

Name of the parameter	Value
Partner number	Created logical system
Partner type	LS
Message type	CONF32
Transaction code	CON5

Maintenance of the SAP distribution model - outbound processing

Name of the parameter	Value
To download PM/ CS orders	
Model view	Created model view
Sender/ Client	Logical system of the client
Recipient/ Server	Logical system for the recipient system
Message type	OPERA3
Filter	If necessary, maintain the BDE grouping as filter

Name of the parameter	Value
	criterion
To download the upload request	
Model view	Created model view
Sender/ Client	Logical system of the client
Recipient/ Server	Logical system for the recipient system
Message type	REQUI3

Maintenance of the SAP distribution model - inbound processing

Name of the parameter	Value
To upload time tickets	
Model view	Created model view
Sender/ Client	Logical system for the sender system
Recipient/ Server	Logical system of the client
Message type	CONF32

Relevant transactions

Transaction	Meaning	Please note:
CI32	Initial download	
CI34	Download maintenance/ service orders as delta download	-
CI35	Download of the upload request	-
IW46	Reworking of incorrect postings	-

